

# Web Datamining and Semantics Project

*Knowledge Graph Construction, Alignment, Reasoning, KGE, and RAG over  
RDF/SPARQL*

## Final Report

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Domain: Bench press, strength training, and biomechanics

### Repository structure summary

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This report follows the required grading structure: Data Acquisition and IE, KB Construction and Alignment, SWRL Reasoning, KGE, RAG over RDF/SPARQL, and Critical Reflection.

## Executive Summary

The goal of this project was to build a complete semantic pipeline starting from web pages and ending with a queryable RDF knowledge base, reasoning rules, knowledge graph embeddings, and a small RAG demo over SPARQL. Our domain is the bench press, with a focus on strength training technique and biomechanics. We chose this domain because it is narrow enough to model clearly, but still rich in entities such as muscles, joints, equipment, technique cues, and biomechanical variables.

The pipeline starts with a controlled web acquisition step. We crawl or locally reload a small set of high-quality seed URLs, clean the HTML into article text, and extract useful subject-relation-object triples with a domain-specific NLP pipeline. Those triples are then converted into RDF, aligned through canonical entity and predicate rules, enriched through SPARQL-style expansion rules, and exported as Turtle files. On top of that, we demonstrate SWRL reasoning, train two KGE models, and implement a lightweight natural-language-to-SPARQL RAG demo.

The final knowledge base is not huge, and we are honest about that limitation. Still, the whole end-to-end pipeline is reproducible: from crawling and information extraction to RDF modeling, reasoning, embedding training, evaluation, and interactive SPARQL-based question answering.

| Part                          | Main output                                               | Status |
|-------------------------------|-----------------------------------------------------------|--------|
| Data Acquisition and IE       | Cleaned JSONL + extracted knowledge CSV                   | Done   |
| KB Construction and Alignment | ontology.ttl, initial_kb.ttl, expanded_kb.ttl, statistics | Done   |
| Reasoning (SWRL)              | family.owl output + KB SWRL inference                     | Done   |
| Knowledge Graph Embeddings    | TransE, DistMult, metrics, t-SNE, nearest neighbors       | Done   |
| RAG over RDF/SPARQL           | Schema summary, NL-to-SPARQL demo, evaluation, screenshot | Done   |

## 1. Data Acquisition and Information Extraction

### 1.1 Domain and Seed URLs

Our domain is bench press technique and biomechanics. We selected seed URLs that mix practical strength training explanations with more scientific or biomechanics-oriented content. The objective was not to crawl the whole web; it was to build a clean, controlled corpus that is directly useful for a semantic knowledge graph.

| Source type                | Seed URL / page                                                  |
|----------------------------|------------------------------------------------------------------|
| Fitness/practical          | Nike - how and why to do a bench press                           |
| Technique/anatomy          | Muscle and Motion - how to bench press properly                  |
| Biomechanics               | NASM - biomechanics of the bench press                           |
| Scientific article         | Frontiers in Sports and Active Living - bench press article      |
| Strength training analysis | Robertson Training Systems - biomechanics and the bench press    |
| Strength training guide    | Stronger by Science - how to bench                               |
| Scientific article         | MDPI article, skipped during live crawl because of 403 Forbidden |

## 1.2 Crawler Design and Ethics

The crawler was designed as a controlled seed-page crawler. It can run in live mode to fetch the seed pages, but it also saves the HTML locally so that later experiments can run from cached files. This makes the pipeline reproducible and avoids repeatedly hitting external websites.

- A custom user-agent is used to identify the crawler as an educational project.
- A delay is applied between live requests to avoid aggressive scraping.
- The crawler only targets manually selected public pages relevant to the project domain.
- If a page returns an access error, such as the MDPI 403 Forbidden response, the crawler skips it instead of trying to bypass the restriction.
- The local HTML mode is the recommended mode for repeated experiments.

## 1.3 Cleaning Pipeline

The cleaning step uses Trafilatura to extract the main readable article text from HTML. It removes most boilerplate such as navigation menus, scripts, comments, and irrelevant page structure. We then reject pages where the extracted text is too short, store word counts and timestamps, and compute a text hash for duplicate detection.

*HTML page -> Trafilatura extraction -> useful-content filter -> text hash -> JSONL cleaned corpus*

## 1.4 NER and Relation Extraction Examples

In this project, NER is domain-specific. Instead of only extracting standard named entities like locations or organizations, we extract exercise concepts, muscles, joints, equipment, technique cues, and biomechanical variables. Dependency parsing is then used to extract relations between those entities.

| Entity type                                           | Examples                                       | Why it matters                                         |
|-------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Muscle                                                | pectoralis major, triceps, anterior deltoid    | Muscles become important nodes in the KG.              |
| Joint / body part                                     | shoulder, elbow, wrist, scapula                | Useful for biomechanics and reasoning.                 |
| Equipment                                             | bar, barbell, bench                            | Links exercise actions to equipment.                   |
| Technique cue                                         | grip width, bar path, scapular retraction      | Useful for RAG questions about technique.              |
| Biomechanical concept                                 | force, moment arm, range of motion, activation | Captures the scientific/semantic part of the domain.   |
| Text / context                                        | Extracted entities                             | Extracted triple                                       |
| The triceps help extend the elbow.                    | triceps, elbow                                 | triceps -> extends -> elbow                            |
| The pectoralis major contributes to shoulder flexion. | pectoralis major, shoulder                     | pectoralis_major -> flexes -> shoulder                 |
| The bar applies a downward force.                     | bar, downward force                            | bar -> appliesForce -> downward_force                  |
| Grip width affects movement demands.                  | grip width, horizontal flexion demands         | grip_width -> influences -> horizontal_flexion_demands |
| Scapular retraction limits scapular movement.         | scapular retraction, scapula                   | scapular_retraction -> reduces -> scapula              |

## 1.5 Ambiguity Cases

We identified three important ambiguity cases during IE:

1. Pecs / chest / pectoralis major: casual sources use “pecs” or “chest”, while scientific sources use “pectoralis major”. We normalize these to pectoralis major to avoid duplicate KG nodes.

2. Bar / barbell / general bar: in this domain, “bar” usually means the barbell, but in general English it has many meanings. We keep it only when the sentence is clearly related to bench press technique or biomechanics.
3. Press as exercise, action, or generic word: “press” can mean bench press, overhead press, leg press, the act of pressing, or even media/press. We only keep it when the context contains domain signals like bar, bench, triceps, shoulder, or force.

## 2. Knowledge Base Construction and Alignment

### 2.1 RDF Modeling Choices

The extracted triples are converted into RDF using the classic subject-predicate-object structure. Each cleaned entity becomes a URI in the project namespace. Predicates are also aligned to controlled RDF properties. We also keep evidence through statement nodes, so we do not lose the original extraction context or confidence score.

*triceps -> extends -> elbow*  
*bar -> appliesForce -> downward\_force*  
*grip\_width -> affectsBiomechanics -> horizontal\_flexion\_demands*

---

### 2.2 Ontology and Entity Classes

The ontology defines classes such as Exercise, Muscle, Joint, Equipment, TechniqueCue, BiomechanicalConcept, DomainEntity, and KnowledgeStatement. This gives the KG a clean structure and allows later modules, especially SPARQL and SWRL, to ask type-aware questions.

| Class                | Example instances                                  |
|----------------------|----------------------------------------------------|
| Exercise             | bench press, incline bench press                   |
| Muscle               | triceps, pectoralis major, anterior deltoid        |
| Joint                | elbow, shoulder, wrist, scapula                    |
| Equipment            | bar, bench                                         |
| TechniqueCue         | grip width, bar path, scapular retraction          |
| BiomechanicalConcept | force, range of motion, horizontal flexion demands |

### 2.3 Entity Linking with Confidence

Entity linking maps surface forms to canonical entities. For example, pecs, chest, and pectorals are mapped to pectoralis major. The confidence values produced during extraction are kept in the RDF graph through KnowledgeStatement nodes, together with source files and example evidence sentences. This is important because web extraction is not perfect, so confidence should stay visible.

| Raw mention | Canonical entity |
|-------------|------------------|
| pecs        | pectoralis major |
| chest       | pectoralis major |
| pectorals   | pectoralis major |
| barbell     | bar              |
| scapulae    | scapula          |

## 2.4 Predicate Alignment

Raw extracted predicates are noisy because different texts use different verbs for the same idea. We aligned these to a smaller controlled predicate vocabulary. This makes the graph easier to query and reduces relation sparsity.

| Extracted relation variants          | Aligned predicate   |
|--------------------------------------|---------------------|
| activate, target, train, work        | targetsMuscle       |
| affect, influence, determine, change | influences          |
| apply                                | appliesForce        |
| extend                               | extends             |
| flex                                 | flexes              |
| require, involve                     | requires / involves |
| reduce, limit                        | reduces             |

## 2.5 Expansion Strategy and KB Statistics

After building the initial graph, we expanded it using SPARQL CONSTRUCT-style rules. The expansion was intentionally conservative: the goal was to add useful semantic relations without creating a lot of noisy inferred facts.

- If a muscle flexes or extends a joint, add muscle -> actsOnJoint -> joint.
- If a technique cue affects a biomechanical concept, add cue -> affectsBiomechanics -> concept.
- If equipment applies force, add equipment -> involvesBiomechanicalConcept -> force.
- If two concepts are connected by requires/involves, add relatedTo as a broader relation.

| Statistic            | Initial KB | Expanded KB |
|----------------------|------------|-------------|
| RDF triples          | 2180       | 2188        |
| Entities             | 344        | 347         |
| Predicates           | 83         | 86          |
| Classes              | 8          | 8           |
| Knowledge statements | 104        | 104         |
| Average confidence   | 0.4361     | 0.4361      |

*The expansion added 8 triples, 3 entities, and 3 predicates. This is small, but it is controlled and avoids over-expanding noisy extractions.*

## 3. Reasoning with SWRL

The reasoning part demonstrates how explicit RDF/OWL facts can be used to infer new facts with SWRL rules. We implemented two examples: a standard family ontology rule and one rule on our bench press KB.

### 3.1 Family Ontology Rule

The family ontology contains three individuals and two explicit parent relations:

*John parentOf Mary*  
*Mary parentOf Alice*

---

The SWRL rule is:

*parentOf(?x, ?y) ^ parentOf(?y, ?z) -> grandparentOf(?x, ?z)*

---

After running Pellet through Owlready2, the reasoner inferred:

### 3.2 Bench Press KB Rule

For our own KB, we used a rule based on technique cues and biomechanical concepts:

*TechniqueCue(?cue) ^ BiomechanicalConcept(?concept) ^ affectsBiomechanics(?cue, ?concept)  
-> ImportantTechniqueCue(?cue)*

---

The KB contained the explicit fact:

*grip width affectsBiomechanics horizontal flexion demands*

---

The reasoner inferred:

*grip width rdf:type ImportantTechniqueCue*

---

This is a small rule, but it is useful because it turns a raw graph relation into a higher-level semantic classification: a technique cue that affects biomechanics is important for bench press technique.

## 4. Knowledge Graph Embeddings

### 4.1 Data Cleaning and Splits

For the embedding part, we did not train on the entire RDF file directly. RDF graphs contain metadata such as labels, confidence values, source documents, and evidence sentences. Those are useful for traceability, but not ideal for link prediction. So we removed metadata triples and kept only semantic object-property triples.

| Dataset statistic  | Value |
|--------------------|-------|
| Semantic triples   | 109   |
| Training triples   | 87    |
| Validation triples | 11    |
| Test triples       | 11    |
| Entities           | 151   |
| Relations          | 68    |

### 4.2 Models and Training Configuration

We trained two KGE models: TransE and DistMult. TransE models relations as translations in the embedding space, while DistMult uses a bilinear scoring function. This gives us two different baselines for link prediction.

| Parameter           | Value |
|---------------------|-------|
| Embedding dimension | 64    |
| Epochs              | 250   |
| Batch size          | 64    |
| Learning rate       | 0.01  |
| Margin              | 1.0   |
| Device              | CPU   |

### 4.3 Metrics and Results

The models were evaluated using link prediction metrics: MRR, Hits@1, Hits@3, and Hits@10. The results are not high in absolute value, which is expected for a small and sparse graph, but the pipeline is complete and the metrics are computed correctly.

| Model    | MRR    | Hits@1 | Hits@3 | Hits@10 |
|----------|--------|--------|--------|---------|
| TransE   | 0.0342 | 0.0000 | 0.0000 | 0.0909  |
| DistMult | 0.1051 | 0.0909 | 0.0909 | 0.0909  |

DistMult performed better than TransE on the main full-dataset run. The nearest-neighbor examples and the t-SNE visualization were also generated to inspect the learned embedding space. Some nearest neighbors are meaningful, while others reveal noise from the extraction stage, which is useful feedback for the overall pipeline.

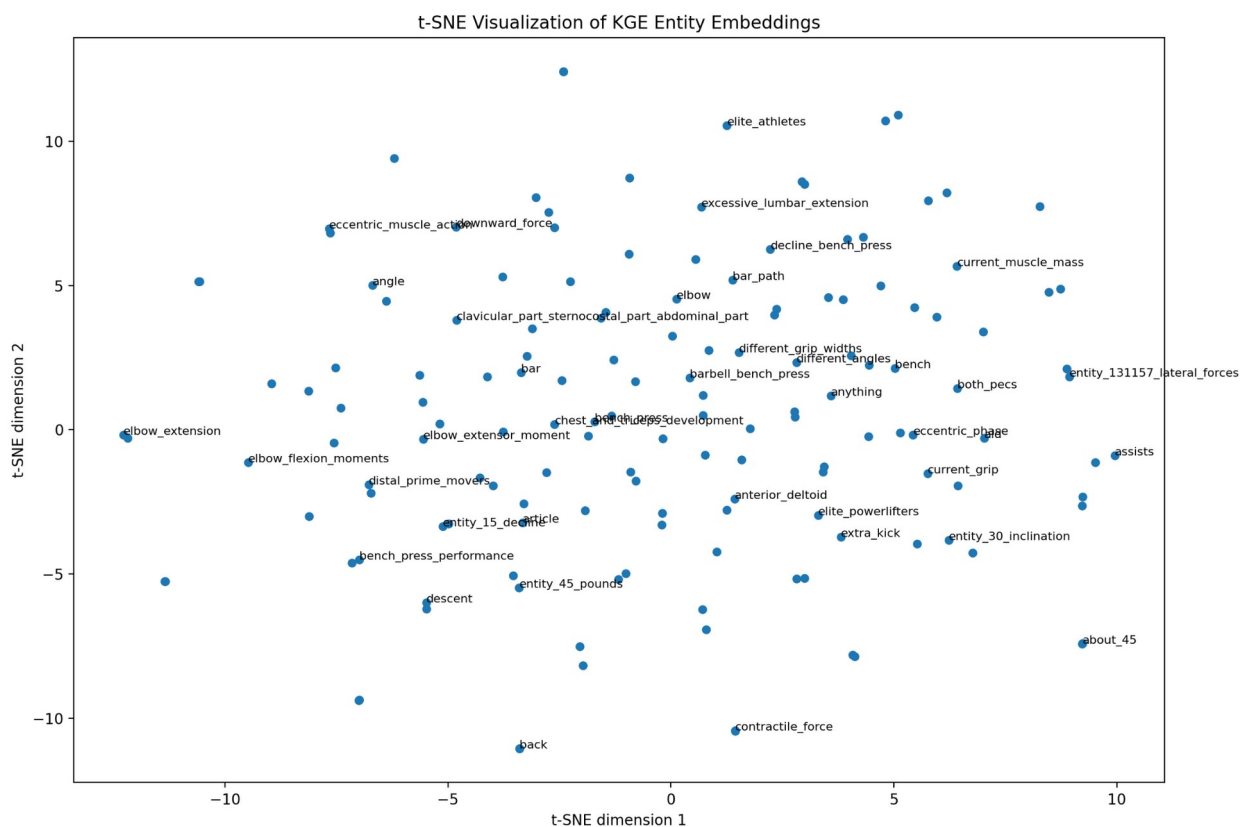


Figure 1. t-SNE visualization of KGE entity embeddings.

## 4.4 Size Sensitivity

The assignment asks for size sensitivity at 20k, 50k, and full graph size. Since our domain-specific KG is much smaller than 20k semantic triples, we created capped subsets: `size_20k_cap`, `size_50k_cap`, and `size_full`. This keeps the experiment honest while preserving the same logic that would scale to a larger graph.

## 5. RAG over RDF/SPARQL

### 5.1 Schema Summary

The RAG module starts by generating a schema summary from the expanded RDF graph. The schema summary lists the namespace, classes, predicates, instance counts, and example triples. This schema is what guides natural-language-to-SPARQL generation.

*PREFIX ex: <http://example.org/benchpress-kg/>*

*Classes: Exercise, Muscle, Joint, Equipment, TechniqueCue, BiomechanicalConcept*

*Predicates: targetsMuscle, actsOnJoint, affectsBiomechanics, appliesForce, extends, flexes, influences, relatedTo*

---

### 5.2 NL-to-SPARQL Prompt Template

The system uses a lightweight rule-based NL-to-SPARQL component. Conceptually, it follows a prompt template: given the user question and the schema, generate a valid SELECT query using only known classes and predicates. We used a rule-based mapper instead of an external LLM to keep the demo reproducible and runnable without API keys.

*Question: Which technique cues affect biomechanical concepts?*

```
SELECT ?cueLabel ?conceptLabel WHERE {  
  ?cue a ex:TechniqueCue .  
  ?concept a ex:BiomechanicalConcept .  
  ?cue ex:affectsBiomechanics ?concept .  
  OPTIONAL { ?cue rdfs:label ?cueLabel . }  
  OPTIONAL { ?concept rdfs:label ?conceptLabel . }  
}
```

---

### 5.3 Self-Repair Mechanism

The RAG demo includes a self-repair mechanism. If the SPARQL query fails, the system attempts to add missing prefixes, repair common predicate casing issues such as appliesforce -> appliesForce, and retry. If the query is valid but returns no result, it falls back to a broader keyword-based SPARQL query over project triples.

### 5.4 Evaluation: Baseline vs RAG

We evaluated at least five natural-language questions. The baseline is a generic keyword-style answer, while the RAG answer is grounded in RDF/SPARQL results from the knowledge base. The RAG answer is more explicit because it returns actual facts from the graph.

| Question                                            | Baseline style                          | RAG/SPARQL answer style                                             |
|-----------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------|
| Which technique cues affect biomechanical concepts? | Grip may affect biomechanics.           | grip width affects horizontal flexion demands                       |
| Which muscles act on joints?                        | Triceps and chest muscles are involved. | triceps actsOnJoint elbow;<br>pectoralis_major actsOnJoint shoulder |
| What force does the bar apply?                      | The bar applies force.                  | bar appliesForce downward_force                                     |
| What does grip width influence?                     | Grip width influences mechanics.        | grip_width influences<br>horizontal_flexion_demands                 |
| Which muscles extend joints?                        | Triceps extend the elbow.               | triceps extends elbow                                               |



```

PS C:\Users\Maxia\Documents\02_ESILV\AI\58\Web_datamining_and_semantics\Web_Datamining_and_Semantics_Project_Maxia_Grossmann_Geoffroy_Gankoue\06_RAG_over_RDF_SPARQL> python .\06b_rag_sparql_demo.py
RAG over RDF/SPARQL Demo
=====
Ask questions about the bench press knowledge base.
Type 'exit' to quit.

Example questions:
- Which technique cues affect biomechanical concepts?
- Which muscles act on joints?
- What force does the bar apply?
- What does grip width influence?
- Which muscles extend joints?
=====

Question: Which technique cues affect biomechanical concepts?
=====
Question: Which technique cues affect biomechanical concepts?
=====

Generated SPARQL:
PREFIX ex: <http://example.org/benchpress-kg/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?cueLabel ?conceptLabel WHERE {
  ?cue a ex:TechniqueCue .
  ?concept a ex:BiomechanicalConcept .
  ?cue ex:affectsBiomechanics ?concept .
  OPTIONAL { ?cue rdfs:label ?cueLabel . }
  OPTIONAL { ?concept rdfs:label ?conceptLabel . }
}

Answer:
Technique cues affecting biomechanical concepts:
- grip width affects horizontal flexion demands

Self-repair used: False
Fallback used: False

```

Figure 2. Terminal demo showing natural-language question, generated SPARQL, RDF answer, and repair status.

## 6. Critical Reflection

### 6.1 Impact of KB Quality

The biggest lesson is that every later step depends strongly on the quality of the extracted knowledge. If the crawler extracts noisy text or the IE step creates weak triples, that noise appears again in the RDF graph, the KGE embeddings, and the RAG answers. The confidence scores and evidence sentences help keep this transparent instead of pretending that all facts are equally reliable.

### 6.2 Noise Issues

Some extracted entities are still noisy, such as numeric fragments or generic words. This showed up clearly in the nearest-neighbor analysis, where some neighbors were meaningful and others were not. This is not a failure of the whole project; it is an honest sign that information extraction from web pages is hard and that cleaning/alignment is a key part of semantic systems.

### 6.3 Rule-Based vs Embedding-Based Reasoning

The rule-based reasoning part is precise and easy to explain. For example, the SWRL rule that turns a technique cue affecting biomechanics into an ImportantTechniqueCue is transparent. The downside is that we have to write the rules manually. The embedding-based reasoning is more flexible and can reveal similarity patterns, but it is harder to interpret and needs a larger, denser graph to perform well. In our project, the two approaches complement each other: rules give controlled logical inference, while embeddings give exploratory structure.

### 6.4 What We Would Improve

- Add more seed pages and crawl a larger controlled corpus to increase graph size and density.
- Improve entity linking with stronger dictionaries or external anatomical resources.
- Use a more advanced relation extraction model instead of only dependency patterns.
- Add confidence thresholds before RDF construction to reduce noise.
- Use an actual LLM for NL-to-SPARQL generation, while keeping the same self-repair mechanism.

- Train KGE models on a larger graph so that 20k / 50k / full size sensitivity becomes more meaningful.

## 6.5 Final Conclusion

Overall, the project successfully integrates the main semantic web pipeline: data acquisition, information extraction, RDF modeling, alignment, expansion, SWRL reasoning, KGE training, and RAG over SPARQL. The graph is small, but the pipeline is complete, reproducible, and honest about its limitations. The strongest parts are the clean modular structure, the traceability of extracted facts, and the final SPARQL-based demo that answers questions directly from the RDF knowledge base.

## Appendix: Main Artifacts

| Artifact                     | Purpose                                      |
|------------------------------|----------------------------------------------|
| 02A_crawl_and_clean.py       | Controlled crawler and HTML-to-text cleaning |
| 02B_extract_entities.py      | NER and domain-specific triple extraction    |
| 03A_build_initial_rdf.py     | Builds ontology.ttl and initial_kb.ttl       |
| 03B_alignment_rules.py       | Entity and predicate alignment rules         |
| 03C_expand_kb.py             | SPARQL-style KB expansion                    |
| 03D_kb_statistics.py         | Final KB statistics                          |
| 04A_family_swrl_reasoning.py | Family ontology SWRL reasoning               |
| 04B_kb_swrl_reasoning.py     | Project KB SWRL reasoning                    |
| 05A_prepare_kge_dataset.py   | KGE data cleaning and splits                 |
| 05B_train_kge_models.py      | TransE and DistMult training/evaluation      |
| 05C_analyze_kge_results.py   | Nearest neighbors and t-SNE                  |
| 06A_schema_summary.py        | RDF schema summary                           |
| 06B_rag_sparql_demo.py       | Interactive RAG over RDF/SPARQL demo         |
| 06C_evaluate_rag.py          | Baseline vs RAG evaluation                   |